

Revealing the mechanism of carrier transport in host-guest systems of organic materials with a modified Poisson and drift-diffusion solver

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A numerical solver based on two-dimensional Poisson and drift-diffusion equations with Gaussian-shaped density of states, the Poole-Frenkel field mobility model, and the Ising model are proposed to analyze carrier transport in the host-guest systems based on organic semiconductors. To investigate the performance of this solver, a series of the electron-only devices and hole-only devices are fabricated with different dopant concentrations. The space-charge-limited current model is applied to the simulation results to calculate the mobility. In addition, experimental results for different host-guest systems are compared to prove the accuracy of the solver. The solver is able to explain the physical mechanisms in different situations. At lower dopant concentrations, the guest states act as traps. Carriers are localized in these guest states, and the carrier mobility decreases dramatically. At the higher dopant concentrations, regions of guest material are able to connect with each other, and carriers have a probability to hop through the guest states. The mobility characteristics gradually approach those of the pure guest material.

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I. INTRODUCTION

Since the first organic light-emitting diodes (OLEDs) were demonstrated by Tang in 1987 [1], they have attracted considerable attention [2–5]. However, OLEDs suffer from limitations on their internal quantum efficiency (IQE). For organic materials, according to the spin selection rule, excitons formed from electron-hole pairs are divided into 25% singlet excitons and 75% triplet excitons. Hence the external quantum efficiency (IQE) of the first-generation OLEDs, namely, fluorescent OLEDs, is quite low. However, the phosphorescent OLEDs (PhOLEDs) and thermally activated delayed fluorescence (TADF) OLEDs, invented by Adachi *et al.* in 2001 and 2012, respectively [6,7], can theoretically achieve IQEs of 100%. These developments made the commercialization of OLEDs feasible. In practice, TADF OLEDs have recently achieved EQEs of more than 30% [8–11]. Some studies have shown that exciton confinement is key to designing high-performance OLEDs [12,13]. Many reports have shown that the position of exciton generation should be controlled to be at the middle of the emissive layer (EML) and that this

position is determined by the mobilities of electrons and holes [14–16]. Hence mobility matching of electrons and holes is one of the most important factors determining the efficiency of organic semiconductors. The EML of a PhOLED usually consists of a mixture of more than two materials, and recent studies have shown that for TADF OLEDs, the EQE of a doped EML is better than that of a nondoped EML [17]. Hence, both PhOLEDs and TADF OLEDs can be classified as host-guest systems. Some studies have shown that the mobility changes in the host-guest system exhibit a nonlinear trend as the dopant concentration varies [18–21]. However, the mechanism underlying these changes cannot be easily analyzed with numerical models. Experimentally, for organic materials, mobility parameters are usually extracted from time-of-flight (TOF) measurements [22,23], which have the advantage that they can be performed with the material subjected to different electric fields. However, this method of measurement has the disadvantage that a thick sample is required, generally of thickness more than 1 μm , which is costly [24–26]. In recent years, some numerical methods have been proposed to calculate carrier transport in blended organic systems [27–31]. Prominent among these is the kinetic Monte Carlo (kMC) method, which is usually used for disordered systems. It has been successfully applied to study organic

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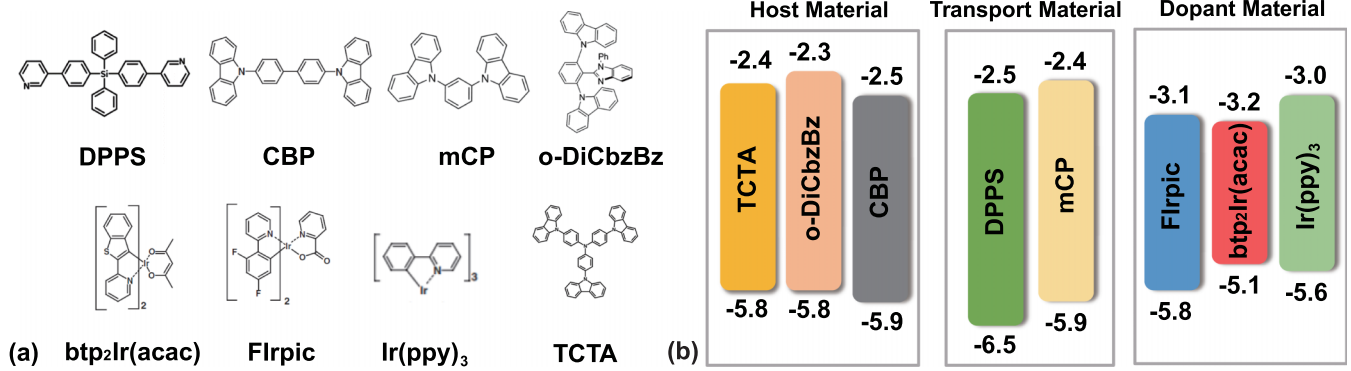


FIG. 1. (a) Molecular structures of DPPS, CBP, mCP, o-DiCzBz, btp₂Ir(acac), Flrpic, Ir(ppy)₃, and TCTA. (b) Schematic representation of the LUMOs and HOMOs of the materials used in this work.

photovoltaics (OPVs) [32–34]. To self-consistently solve the potential and current density, a Poisson and drift-diffusion (Poisson-DD) solver is typically used, which provides an efficient tool to model the current flow and potential variation in the devices under biased operation. Parameters in kMC and Poisson-DD are different, but they have a similar physical meaning (see Supplemental Material Table S1 [35] for the comparison of parameters [29]). It can also include the hole transport layer (HTL), electron transport layer (ETL), and EML in the simulation to model the OLED behavior. However, few studies based on the Poisson-DD method were used to predict carrier transport in the host-guest system with different dopant concentrations [36,37].

In this work, to demonstrate the properties of an organic host-guest system, a modified Poisson-DD solver is established, which includes a Gaussian-shaped density of states (DOS), the Poole-Frenkel field-dependent mobility model, a random dopant model, and the two-dimensional (2D) Ising model. A series of electron-only devices (EODs) and hole-only devices (HODs) with a blended active layer are fabricated to check the accuracy of the solver. In addition, some iconic host-guest systems are simulated for comparison with TOF experimental results in the literature [38,39]. It is found that computations using the proposed solver are much more efficient than those based on the kMC method.

II. METHODOLOGY

The Poisson-DD solver is based on the two-dimensional drift-diffusion charge control solver (2D-DDCC) developed by our laboratory [40]. This software has been utilized in several studies, including those of light-emitting diodes, solar cells, and photodetectors [41–43]. To examine the accuracy of the solver, the EOD and HOD were fabricated with different dopant concentrations. For investigations of changes in mobility in the host-guest system, the host material used was 9,9'-(2-(1-phenyl-1*H*-benzo[*d*]imidazol-2-yl)-1,3-phenylene)bis(9*H*-carbazole) (o-DiCzBz), which was synthesized by the Department of Chemistry, National Taiwan University [44], and the dopant material was bis[2-(4,6-difluorophenyl)pyridinato-C2,N] (picolinate)iridium(III) (Flrpic), and Fig. 1(a) shows the molecular structures of these and other materials used in this study. The difference in lowest unoccupied molecular orbital (LUMO) between

dopant and host material is 0.8 eV, and the highest occupied molecular orbitals (HOMOs) are the same. Figure 1(b) shows a schematic representation of the HOMOs and LUMOs of these and other materials used in this study. Hence this system provides a good example for understanding clearly the mechanisms involved in the host-guest system in different situations, including both deep and shallow energy gaps. The band diagrams of the EOD and HOD are shown in Figs. 2(a) and 2(b). For the EOD, diphenyl-bis[4-(pyridin-3-yl)phenyl]silane (DPPS) was used as ETL and Al/LiF as the cathode. For the HOD, *N,N'*-dicarbazolyl-3,5-benzene (mCP) was used as the HTL and Al/MoO₃ as the anode. For both the EOD and HOD, the host material was o-DiCzBz and the guest material was Flrpic. Furthermore, different host-guest systems were also investigated, with o-DiCzBz, 4,4'-bis(*N*-carbazolyl)-1,1'-biphenyl (CBP), and

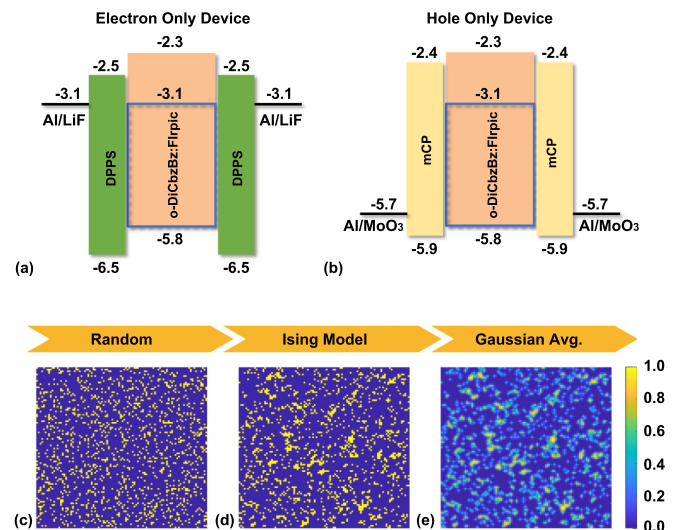


FIG. 2. (a, b) Schematic structures of the EOD and HOD. (c) The initial map is generated by a random process in which 1 and 0 are host and guest, respectively. (d) The map is assigned by the 2D Ising model. (e) In the Gaussian-averaging map, a value between 0 and 1 means that the carrier can transport in a substrate of the guest, and the DOS is modified accordingly. In this case, the dopant concentration is 20%, and the summations of the whole map are the same for (c), (d), and (e).

TABLE I. Parameters of the density of states used in this work.

Material	LUMO	HOMO	$N_{t,e}$ (cm ⁻³)	$E_{t,e}$ (eV)	$\sigma_{t,e}$ (eV)	$N_{t,h}$ (cm ⁻³)	$E_{t,h}$ (eV)	$\sigma_{t,h}$ (eV)
colrule o-DiCzBz	2.3	5.8	1.0×10^{21}	2.3	0.13	1.0×10^{21}	5.1	0.10
CBP	2.5	5.9	1.0×10^{21}	2.5	0.13	1.0×10^{21}	5.9	0.11
TCTA	2.5	5.8	1.0×10^{21}	—	—	1.0×10^{21}	5.8	0.07
mCP	2.4	5.9	1.0×10^{21}	—	—	1.0×10^{21}	5.9	0.15
DPPS	2.5	6.5	1.0×10^{21}	2.5	0.12	1.0×10^{21}	—	—
FIrpic	3.1	5.8	1.0×10^{21}	3.4	0.12	1.0×10^{21}	5.8	0.12
bt ₂ Ir(acac)	3.2	5.1	1.0×10^{21}	3.0	0.11	1.0×10^{21}	—	—
Ir(ppy) ₃	3.0	5.6	1.0×10^{21}	—	—	1.0×10^{21}	5.6	0.14

tris(4-carbazoyl-9-ylphenyl)amine (TCTA) as host materials, and FIrpic, bt₂Ir(acac), and Ir(ppy)₃ as guest materials. The parameters of the DOS and mobility are shown in Tables I and II, respectively. A recent study has shown that the DOS of organic materials is related to their absorption spectra [45]. Hence the DOS parameters refer to the absorption spectra (see Supplemental Material Fig. S1 [35] for absorption spectra of all the materials used in this work [38,44,46–49]). The mobility parameters are obtained from reported TOF measurements, as indicated in Table II. However, no published TOF data are available for DPPS, and so EOD devices of different thicknesses were fabricated to estimate the mobility parameters of this material (see Supplemental Material Fig. S2 for the results [35]).

A. Experimental

For device fabrication, all the organic materials were deposited by thermal evaporation in a high-vacuum chamber with pressure less than 8×10^{-6} Torr and with the rate of deposition controlled at 1 Å/s. The detailed EOD structure was glass substrate/Al (50 nm)/LiF (1.5 nm)/DPPS (10 nm)/o-DiCzBz:FIrpic (0, 3, 9, 12, 15, 18, 20, 30, and 50 vol %, respectively, 100 nm)/LiF (1.5 nm)/Al (100 nm), and the detailed HOD structure was glass substrate/Al (50 nm)/MoO₃ (10 nm)/mCP (10 nm)/o-DiCzBz:FIrpic (0, 5, 10, 15, 20, 30, and 50 vol %, respectively, 100 nm)/mCP (10 nm)/MoO₃ (10 nm)/Al (100 nm). The device characteristics, including current density and voltage, were measured by a brightness–current density–voltage (BIV) system. A Keithley 2400 electrometer was used to record current density and voltage, and the system was controlled by LABVIEW software.

B. Simulation

1. Modified Poisson and drift-diffusion equation

The two-dimensional finite-element Poisson-DD solver is utilized to model the characteristics of organic materials. Briefly, this solver is based on a self-consistent solution of the Poisson equation,

$$\nabla_r \cdot [\varepsilon \nabla_r V(r)] = q(n - p + \dots), \quad (1)$$

the drift-diffusion equations,

$$J_n = -q\mu_n n(r) \nabla_r V(r) + qD_n \nabla_r n(r), \quad (2)$$

$$J_p = -q\mu_p p(r) \nabla_r V(r) - qD_p \nabla_r p(r), \quad (3)$$

and the continuity equation

$$\nabla_r(J_{n,p}) = q(R - G), \quad (4)$$

where q is the elementary charge, ε is the dielectric constant, V is the potential, n and p are the electron and hole densities, respectively, J_n and J_p are the electron and hole current densities, μ_n and μ_p are the electron and hole mobilities, D_n and D_p are the electron and hole diffusion coefficients, and R and G are the recombination rate and generation rate, respectively.

Several studies have shown that the distribution of DOS for organic materials is different from that characteristic of inorganic semiconductors [50,51]. For organic materials, carrier transport can occur in the tail energy states (below LUMO and above HOMO) by a hopping process, which can be represented by a combination of multi-Gaussian distribution functions. To describe carrier transport in organic materials, the following Gaussian-shaped DOS function is introduced to

TABLE II. Mobility parameters used in this work.

Mobility	$\mu_{0,e}$	β_e	$\mu_{0,h}$	β_h
	[cm ² /(V s)]	[(cm/V) ^{1/2}]	[cm ² /(V s)]	[(cm/V) ^{1/2}]
o-DiCzBz (Ref. [44])	4.0×10^{-8}	5.9×10^{-3}	2.0×10^{-8}	9.3×10^{-3}
CBP (Ref. [38])	4.1×10^{-4}	7.0×10^{-4}	3.9×10^{-4}	2.7×10^{-3}
TCTA (Ref. [39])	—	—	9.4×10^{-5}	1.3×10^{-3}
mCP (Ref. [59])	3.9×10^{-5}	4.3×10^{-3}	1.2×10^{-4}	2.5×10^{-3}
DPPS	4.0×10^{-7}	8.2×10^{-3}	—	—
FIrpic (Ref. [38])	1.0×10^{-9}	9.1×10^{-3}	2.2×10^{-9}	7.5×10^{-3}
bt ₂ Ir(acac) (Ref. [38])	1.8×10^{-4}	1.0×10^{-3}	1.6×10^{-6}	5.4×10^{-3}
Ir(ppy) ₃ (Ref. [20])	—	—	5.0×10^{-6}	4.1×10^{-3}

describe the tail states:

$$N_{\text{tail}}(E) = N_t \frac{1}{\sigma \sqrt{2\pi}} \exp\left[-\frac{(E - E_t)^2}{2\sigma^2}\right], \quad (5)$$

where N_t denotes the total available state for this Gaussian distribution, E_t is the center of the Gaussian distribution, which is typically near the band edge, and σ is a broadening factor. When the Gaussian-shaped DOS is applied, the carrier density can be expressed as

$$n, p = \int_{-\infty}^{\infty} N_{\text{tail}}(E) f_{e,h}(E) dE, \quad (6)$$

where f_e and f_h are the Fermi-Dirac distribution functions for electrons and holes.

Furthermore, the carriers are accelerated by an applied electric field. Hence the mobility behavior can be described by the Poole-Frenkel mobility [52–55], which can be expressed as follows:

$$\mu_{e,h} = \mu_{0,e,h} \exp(\beta_{e,h} \sqrt{|\vec{F}|}), \quad (7)$$

where μ_e and μ_h are the carrier mobilities for electrons and holes, $\mu_{0,e,h}$ are the zero-field mobilities for electrons and holes, $\beta_{e,h}$ are the field activation parameters for electrons and holes, and $\vec{F}$ is the electric field.

2. Space-charge-limited current model

The space-charge-limited current (SCLC) model is a common method to extract the mobility carriers in devices based on organic materials [56–58]. In our modeling, the SCLC model is utilized to extract the mobility at different dopant concentrations to elucidate the physical mechanisms at these concentrations. The J - V curve can be divided into two regions at lower and higher voltages: Ohmic and SCLC regions, respectively. In the Ohmic region, the characteristics of the J - V curve are affected by contact resistance and traps. In the SCLC region, the logarithm of the J - V curve shows a linear trend, with the slope indicating the characteristics of the material in the absence of interference. Hence the mobility parameters can be extracted at high electric fields [$>6.4 \times 10^5$ V/cm], as described in the following sections.

The SCLC model can be expressed as follows:

$$J = \frac{9}{8} \epsilon \mu \frac{V_{\text{bi}}^2}{L^3}, \quad (8)$$

where J is the current density at a specific bias, ϵ is the dielectric constant, μ is the carrier mobility, V_{bi} is the applied bias, and the L is the length of the whole OLED.

3. Ising model

Some studies have shown that blended organic systems are not entirely random systems and that molecular interactions need to be taken into account [60]. The Ising model is a commonly used method to describe the impact of molecular interactions on blended systems, in particular, blended organic systems [29,32,34]. Hence the Ising model is implemented in this work to generate the simulation map, and two dimensions are considered, which are the x and y directions. First, a cubic grid is generated with a size of $10 \text{ \AA}$, which is the average iridium-iridium distance in iridium compounds (see

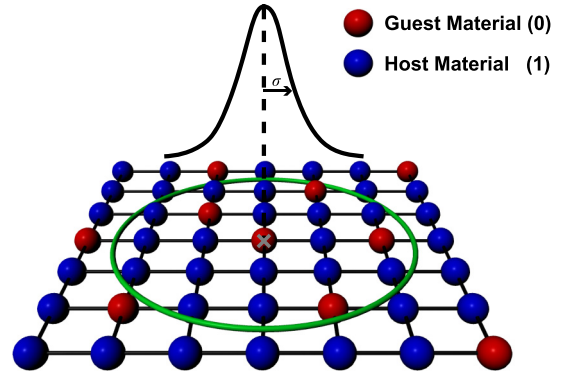


FIG. 3. Schematic representation of Gaussian averaging, in which the distributions of host and guest are decided by a random process and the Ising model.

Supplemental Material Fig. S3 [35] for the crystal structures [61,62]). As shown in Fig. 2(c), the site of the molecule is randomly assigned as either 0 or 1, with 0 and 1 representing host and guest materials, respectively. Exchange dynamics are implemented to relax the system to a lower energy [63], adjacent pairs of host and guest are chosen randomly in the system, and their elements can exchange with a probability that is determined by the energy change in the system, $\Delta\epsilon$. This energy change can be calculated from

$$\epsilon_i = \frac{-J_{\text{Ising}}}{2} \sum_j (\delta_{i,j} - 1), \quad (9)$$

where $\delta_{i,j}$ is the Kronecker delta function. The exchange probability can then be expressed as

$$P(\Delta\epsilon) = \frac{\exp(-\Delta\epsilon/k_B T)}{1 + \exp(-\Delta\epsilon/k_B T)}, \quad (10)$$

where J_{Ising} is the interaction energy, which is generally equal to $k_B T$, where k_B is Boltzmann's constant and T is temperature. Figure 2(d) shows how the guest material clusters and forms into grains.

4. Gaussian averaging method

To describe carrier transport in the host-guest system, the Gaussian-averaging method is applied to demonstrate the hopping process in the tail state in organic materials. The Gaussian-averaging method is illustrated in Fig. 3. For each guest, the substrate is determined from the Gaussian averaging method as follows:

$$\zeta^\dagger(x, y) = \frac{\sum_{i=1}^{N_x} \sum_{j=1}^{N_y} \zeta(x, y) \exp\left[-\frac{(x-i)^2 + (y-j)^2}{2\sigma^2}\right]}{\sum_{i=1}^{N_x} \sum_{j=1}^{N_y} \exp\left[-\frac{(x-i)^2 + (y-j)^2}{2\sigma^2}\right]}, \quad (11)$$

where $\zeta^\dagger$ are the elements of the host-guest system after application of the Gaussian averaging method, and ζ are the initial elements of the host-guest system, N_x and N_y are the maximum nodes in the x and y directions, and σ is the broadening factor in the Gaussian function.

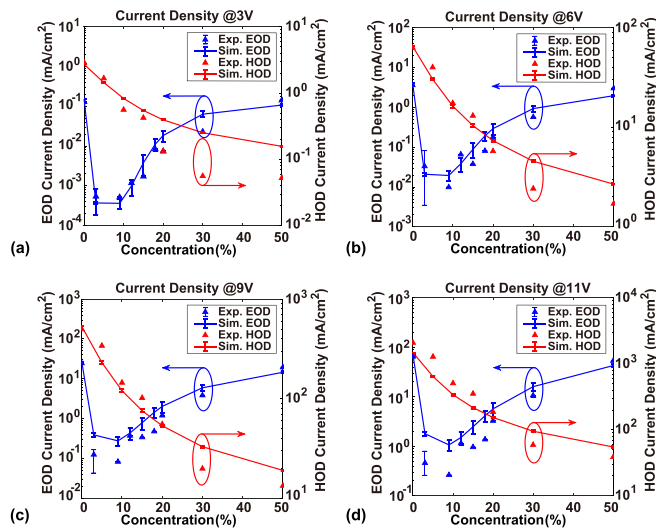


FIG. 4. The current density vs different applied voltages (3, 6, 9, and 11 V) for EOD/HOD with different dopant concentrations for o-DiCzBz:FIrpic.

Finally, the simulation map changes from Fig. 2(d) to Fig. 2(e). A value between 0 and 1 means that the carrier can transport in a substrate of the guest material, and the DOS is modified accordingly. In this work, the simulation area was set as 100×100 nm, which cannot represent the real situation. Hence, 100 different seeding maps are adopted to obtain a much more convergent result, with an error less than 2%. The modeling results from this work include the average and the standard deviation (see Supplemental Material Fig. S4 [35] for the detailed information about the number of seeding maps).

By using this method we can solve the characteristics of organic material more efficiently. For typical 500×500 grid maps (lengths of x direction and y direction are both 100 nm, the grid spacing is 0.2 nm), the converged results can be obtained by 5–10 minutes for a bias point with Intel Xeon E5-2620v4 CPU and four threads.

In addition, some studies [64] suggested that charge-charge interaction resulting from Coulomb interaction is quite important when the carrier density is high. In this case the carrier density is relatively lower and the fitting to the experimental works well. In the future, the carrier-density-dependent mobility model can be added to include this effect.

III. RESULTS AND DISCUSSION

A. Modeling of o-DiCzBz:FIrpic

First, the accuracy of the transport model including random doping effects as shown in Fig. 2 will be examined. The DOS and mobility parameters are shown in Tables I and II, respectively. EODs and HODs were fabricated with EMLs with different dopant concentrations to examine the effect of concentration on the results of the model. Using the 2D random dopant model at different concentrations, the J - V curve could be obtained. Figures 4(a)–4(d) show comparisons between the experimental and simulation results at 3, 5, 8, and 11 V, respectively (see Supplemental Material Fig. S5 [35] for more data of the J - V curve). It can be seen that the experi-

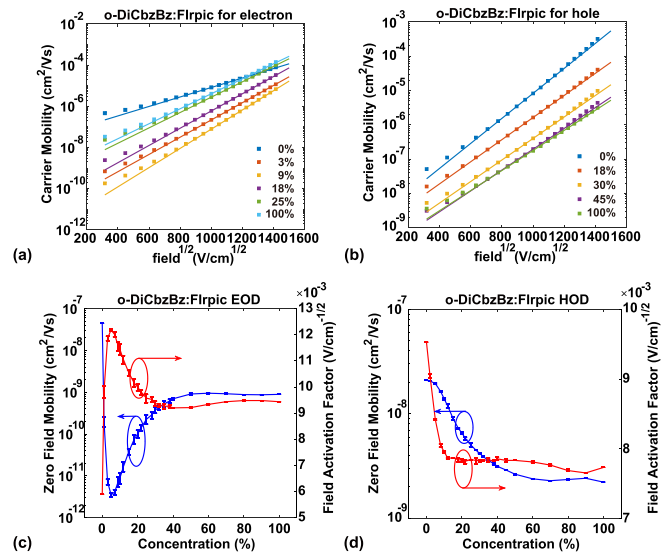


FIG. 5. (a, b) Electron/hole mobility vs electric field for o-DiCzBz:FIrpic with different dopant concentrations. The symbols are modeling results, and the lines are SCLC fitting results. (c, d) Zero-field mobility and field activation parameter at different dopant concentrations for electrons and holes in o-DiCzBz:FIrpic.

mental and simulation results show almost the same trends. At low dopant concentration, the current density decreases with increasing dopant concentration for the EOD (blue lines and dots). When the dopant concentration is greater than 10%, the current density increases gradually. For the HOD (red lines and dots), the current density changes linearly as the dopant concentration increases from 0% to 50%. In this host-guest system, the LUMO of FIrpic (-3.1 eV) is lower than that of o-DiCzBz (-2.3 eV), and the HOMO of FIrpic (-5.8 eV) is the same as that of o-DiCzBz (-5.8 eV). Hence the EOD and HOD can be classified as having deep and shallow energy differences, respectively.

To investigate and study the properties of carrier transport, the HTL and ETL are removed from the modeling structure so that their influences can be ignored. In addition, the contact work function is set the same as the LUMO or HOMO of the material, which is an assumption of the SCLC model [56,58]. Different biases are then applied, and the J - V curve is calculated.

As shown in Figs. 5(a) and 5(b), mobilities at different dopant concentrations and under different electric fields can be obtained by using the SCLC model, and the zero-field mobility and field activation factor can be extracted. Figures 5(c) and 5(d) show the modeling results for the extracted effective mobilities of electrons and holes, respectively. A trend similar to that in Fig. 4 can be observed. The electron mobility initially decreases dramatically as the dopant concentration increases, but when the dopant concentration becomes greater than 10%, the mobility gradually increases again. In fact, many studies have shown the same phenomenon [20,38,65]. The reason for this nonlinear behavior is that the guest state acts as a trap state since the LUMO difference between guest (-3.1 eV) and host (-2.3 eV) is larger at low dopant concentration. When the dopant concentration is higher, the guest

state forms a channel and dominates the blended system. Hence the mobility returns to a plateau region. This means that the guest state will act either as a trap or a channel, depending on whether the dopant concentration is low or high. The mobility of holes changes in a linear manner for dopant concentrations from 0% to 50%. Because the HOMOs of the guest and host are equal (-5.8 eV), the carrier can transport arbitrarily in the host and guest states. In this situation the guest state will not act as a trap, and the nonlinear behavior will not be observed. The mobility depends directly on the doping concentration. It is remarkable that the zero-field mobility and the field activation factor exhibit opposite trends in Figs. 5(c) and 5(d). Some other studies have shown similar results [21,66], and this phenomenon can be explained by the difficulty experienced by the carrier in jumping from guest to host. As a consequence, the field activation factor increases with decreasing zero-field mobility, and vice versa.

B. Modeling of iridium compounds

To further investigate the proposed solver, we applied it to different host-guest systems with CBP and TCTA as host materials and FIrpic, $\text{btp}_2\text{Ir}(\text{acac})$, and $\text{Ir}(\text{ppy})_3$ as guest materials. The mobility data for these systems were obtained by the TOF method [38,39]. Three different cases of the host-guest system are considered: holes in CBP:FIrpic, holes in TCTA: $\text{Ir}(\text{ppy})_3$, and electrons in CBP: $\text{btp}_2\text{Ir}(\text{acac})$. From the band diagrams in Fig. 1(a), it can be seen that the HOMOs of CBP and FIrpic are both -5.8 eV, and the HOMOs of TCTA and $\text{Ir}(\text{ppy})_3$ are -5.8 and -5.6 eV, respectively. The LUMOs of CBP and $\text{btp}_2\text{Ir}(\text{acac})$ are -2.5 and -3.2 eV, respectively. These cases correspond to the situations of no trap, a shallow trap, and a deep trap, respectively. The simulation and experimental results are shown in Fig. 6. The modified Poisson-DD solver can successfully model the trend of mobility variation versus doping concentration. From the results for CBP:FIrpic in Fig. 6(a), it can be seen that the carrier mobility decreases slowly as the dopant concentration increases. In this case the electric field is 3.6×10^5 V/cm, which is the same as the experimental result [38]. Figure 6(b) shows that the SCLC fitting results shift along the vertical direction as increasing dopant concentration and the slopes are almost the same. The slope represents the field activation factor in the SCLC model. In this case there is no barrier in this blended system, but the carrier mobility still decreases slowly. The reason for this is that the mobility of CBP is higher than that of FIrpic (1.6×10^{-3} and 1.5×10^{-7} $\text{cm}^2/(\text{V s})$, respectively). Hence the modeling result approaches the mobility of pure FIrpic gradually as the dopant concentration increases. For TCTA: $\text{Ir}(\text{ppy})_3$, because the HOMO of $\text{Ir}(\text{ppy})_3$ is lower than the HOMO of TCTA, the guest material acts as a shallow trap in this blended system. In this case the electric field is 2×10^5 V/cm, which corresponds to the experimental result [39]. In Fig. 6(a) the mobility decreases from 10^{-4} to 10^{-7} $\text{cm}^2/(\text{V s})$. The hole mobilities of pure TCTA and $\text{Ir}(\text{ppy})_3$ are 1.5×10^{-4} and 2.5×10^{-5} $\text{cm}^2/(\text{V s})$, respectively. However, Fig. 6(a) shows that the mobility with 2% dopant is less than the mobilities of both pure TCTA and $\text{Ir}(\text{ppy})_3$. Hence the reason for the mobility dip at low dopant concentration (near 7%) is the barrier difference in LUMO. For CBP: $\text{btp}_2\text{Ir}(\text{acac})$, the electric field

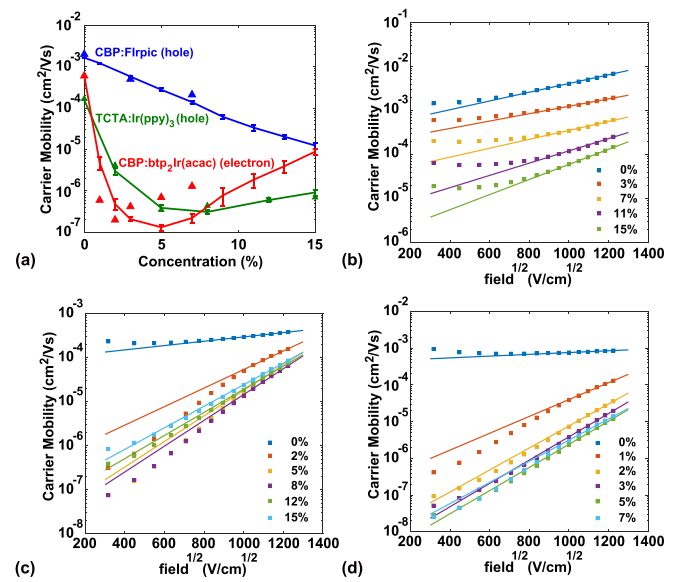


FIG. 6. (a) Carrier mobility for different host-guest systems: CBP:FIrpic ($E = 3.6 \times 10^5$ V/cm), TCTA: $\text{Ir}(\text{ppy})_3$ ($E = 2.0 \times 10^5$ V/cm), and CBP: $\text{btp}_2\text{Ir}(\text{acac})$ ($E = 3.6 \times 10^5$ V/cm). (b)–(d) Zero-field carrier mobility and field activation parameter at different dopant concentrations for holes in CBP:FIrpic, holes in TCTA: $\text{Ir}(\text{ppy})_3$, and electrons in CBP: $\text{btp}_2\text{Ir}(\text{acac})$, respectively. The symbols are modeling results, and the lines are SCLC fitting results.

is 3.6×10^5 V/cm. The LUMO of $\text{btp}_2\text{Ir}(\text{acac})$ is much less than that of CBP. Hence the result in this case is similar to that for TCTA: $\text{Ir}(\text{ppy})_3$. The mobility drops off dramatically after the host (CBP) has been doped with the guest [$\text{btp}_2\text{Ir}(\text{acac})$]. The dip is much larger than in the case of TCTA: $\text{Ir}(\text{ppy})_3$. The reason for this is that the LUMO differential ΔLUMO of $\text{btp}_2\text{Ir}(\text{acac})$ is greater than that of TCTA: $\text{Ir}(\text{ppy})_3$: 0.7 and 0.3 eV, respectively. Next we examine the effect of different combinations of mobility of host and guest and the effect of ΔLUMO to help understand the carrier transport mechanism in organic host-guest systems.

C. Modeling of different conditions

Different host-guest systems with no trap, a shallow trap, and a deep trap have been investigated, and the modeling results show a good fit with experimental results. To understand the mechanisms involved more clearly, we simulate different mobilities of host and guest and different LUMO differentials ΔLUMO . Figure 7(a) shows results for guest mobilities 10^{-4} , 10^{-5} , and 10^{-6} $\text{cm}^2/(\text{V s})$. For all cases in this section, the bias is 4 V and the electric field is 4×10^5 V/cm. For this system the carrier mobility changes monotonically as the dopant concentration increases from 0% to 100%. The guest comes to dominate the system as the dopant concentration gradually increases. In both Figs. 7(b) and 7(c), the dopant concentrations are 25%. Figure 7(b) shows that the carrier concentrates in the guest states because the mobility of the guest is lower than that of the host: 10^{-5} and 10^{-4} $\text{cm}^2/(\text{V s})$, respectively. Figure 7(c) reveals a dramatic concentration of the carrier in guest states when the guest mobility falls to 10^{-6} $\text{cm}^2/(\text{V s})$. Since the mobility in the guest system is

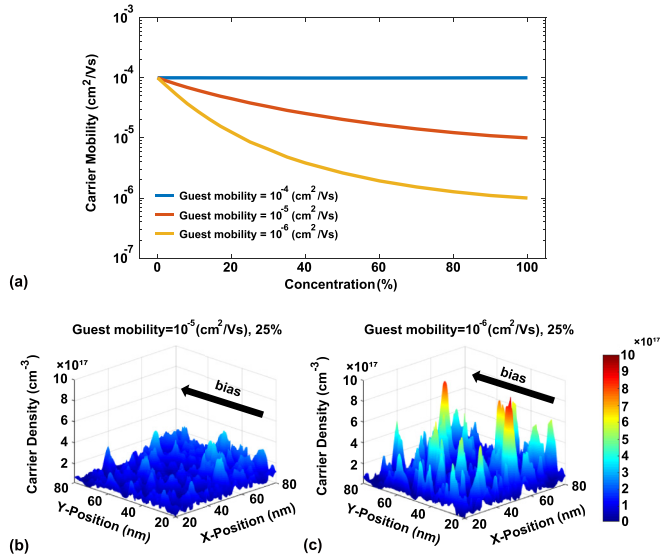


FIG. 7. (a) Carrier mobility for ΔLUMO fixed at 0.0 eV. (b, c) Superpositions of carrier density and host-guest distribution when the mobility of the guest is 10^{-5} and $10^{-6} \text{ cm}^2/(\text{V s})$, respectively. The bias is 4 V, and the electric field is $4 \times 10^5 \text{ V/cm}$.

two orders of magnitude lower, the carrier density needs to be much higher to ensure that the current is continuous through

the whole junction. Hence the carrier is distributed mainly in guest states in this system. However, as shown in Fig. 7(a), the trap effect does not occur, no matter how low the mobility of the guest. This kind of host-guest system corresponds to the holes of o-DiCbzBz:FIrpic and of CBP:FIrpic. Based on the above results, it can be concluded that the mobility of the guest is not the main reason for the dip behavior at low dopant concentration.

We now discuss the influence of ΔLUMO . The results are shown in Fig. 8(a). In this case the mobilities of both host and guest are fixed at $10^{-4} \text{ cm}^2/(\text{V s})$, and ΔLUMO takes values of 0.0, 0.2, 0.4, 0.6, and 0.8 eV. Dip behavior can be clearly observed at low doping concentrations from 1% to 20%. Figures 8(b)–8(e) are schematic representations of carrier density at 10% dopant concentration. ΔLUMO is 0.2 eV in Figs. 8(b) and 8(d), which corresponds to a shallow trap, and 0.8 eV in Figs. 8(c) and 8(e), which corresponds to a deep trap. The bias direction is from bottom to top along the Y direction. Hence electron drift is mainly from top to bottom, which is the opposite direction to the bias. In Figs. 8(b) and 8(c), the blue arrows show the routes of electron transport. Owing to the low dopant concentration, the guest material is isolated. Hence when carriers pass through these localized guest states (either single or clustered guest states), they can easily be trapped and it is difficult for them to jump out. For the grained guest states, although carrier transport is possible,

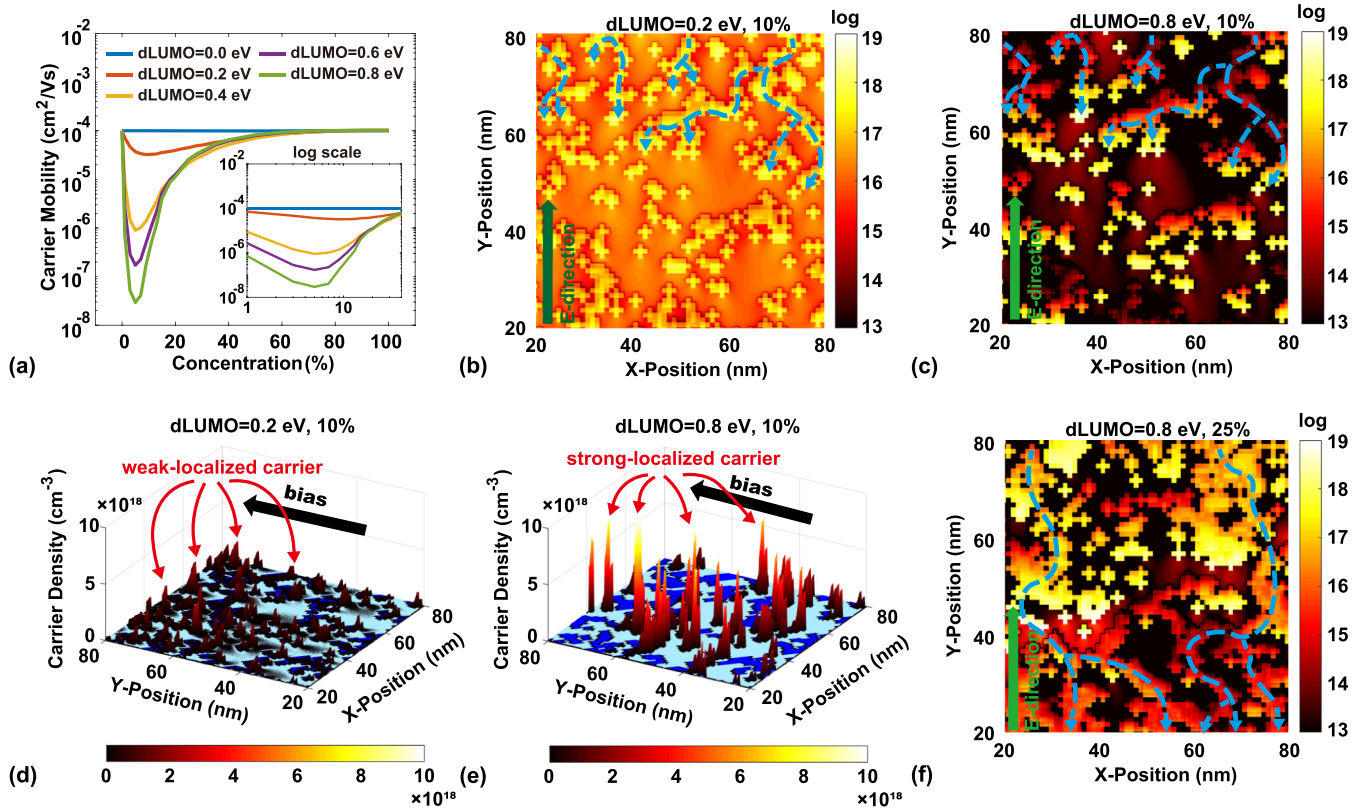


FIG. 8. (a) Carrier mobility for different values of ΔLUMO . (b, c) Superpositions of carrier density and host-guest distribution in a vertical view for $\Delta\text{LUMO} = 0.2$ and 0.8 eV , respectively. (d, e) Superpositions of carrier density and host-guest distribution in a rotated view for $\Delta\text{LUMO} = 0.2$ and 0.8 eV , respectively. (f) Carrier distribution and distribution of host-guest states for $\Delta\text{LUMO} = 0.8 \text{ eV}$ and dopant concentration 25%. For all cases in this figure, the bias is 4 V and the electric field is $4 \times 10^5 \text{ V/cm}$. The regions in deep blue and pale blue denote the guest and host, respectively.

channels cannot form from top to bottom owing to the low dopant concentration. Hence the carriers are mostly confined to these grained guest states, which leads to low mobility. In Figs. 8(b) and 8(d), ΔLUMO (0.2 eV) is small, and as can be seen in Fig. 8(b), although some of the carriers are still localized, the number of localized carriers is less than in the case of the deep trap state (0.8 eV) with the same host-guest distributions [weakly localized carriers can be observed in Fig. 8(d)]. Therefore the carrier density distribution is smoother and the mobility does not decrease dramatically as in the case of $\Delta\text{LUMO} = 0.8$ eV. As can be seen in Fig. 8(a), the mobility gradually approaches 10^{-4} cm²/(V s) when the dopant concentration increases above 25%. The reason for this is that it is easier for the guest states to form a channel from the top contact to the bottom contact, as shown in Fig. 8(f) for 25% dopant concentration. Therefore electrons can flow much more smoothly through these channels. Hence the mobility increases gradually when the dopant concentration becomes greater than 25%.

Furthermore, Fig. 8(a) shows that the turning points of the mobility are different for different values of ΔLUMO . These turning points are at dopant concentrations of 10%, 7%, 5%, and 5% for $\Delta\text{LUMO} = 0.2, 0.4, 0.6$, and 0.8 eV, respectively. This can explain the conflicting results for turning points in the literature [38,39]. In fact, we can observe this result in Fig. 6, where the turning point for TCTA:Ir(ppy)₃ is at a greater dopant concentration than that for CBP:btp₂Ir(acac), with both the simulation and experimental results showing the same trend. The dip in mobility is mainly caused by ΔLUMO , and the value of ΔLUMO affects the depth of the dip.

In this work we can calculate the characteristics of organic semiconductor fitting with experimental results well by several models that we mentioned before. However, some studies have shown that the molecular orientation can be changed by different processes, resulting from variable morphology or crystallinity [67,68], and this leads to changes of mobility and characteristics of electronic properties for organic molecular. But this phenomenon is out of the scope of this manuscript, and since we did not observe this effect in our case, we will discuss this phenomenon in future work.

IV. CONCLUSION

This paper has presented a modified-Poisson and drift-diffusion solver that incorporates the Ising model to consider molecular interactions and generate a simulation map corresponding to the distribution of host and guest materials in real organic host-guest systems. Furthermore, to investigate the performance of this solver, an electron-only device and hole-only device with the emissive layers with different

dopant concentrations have been fabricated. The solver has been combined with an space-charge-limited current model to calculate mobility characteristics, namely, zero-field mobility and the field-activation factor. The results obtained are primarily for an o-DiCbzBz:FIrpic host-guest system. However, by modeling other host-guest systems, namely, CBP:FIrpic, CBP:btp₂Ir(acac), and TCTA:Ir(ppy)₃, the results of the solver have been validated using experimental results from time-of-flight measurements. It is found that three situations can arise in a guest-host system. In the first of these, the guest states act as traps when the energy of the guest is lower than that of the host and the dopant concentration is low. Carriers are then localized in guest states and cannot escape and jump to host states. Hence the carrier mobility drops off dramatically in this situation. The second situation occurs when the dopant concentration is greater than 20%, and the guest states gradually form a channel. Hence the carrier mobility increases and approaches that of the guest. In the third situation the energy states are the same, the carriers can transport arbitrarily in host states and guest states, and there is no barrier between guest states and host states. Hence the impact of mobility is nearly linear, as the dopant concentration varies from 0% to 100%.

See the Supplemental Material Ref. [35] for other information including absorption spectrum for all materials, J-V results of EOD and HOD for DPPs devices, crystal structure for Iridium compounds, number of seeding maps and more data of the J-V curve. The modeling tool is developed by our laboratory and more detailed information is shown on our website [69].

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